

OSSAMA ELHAKKI

Backend & AI Engineer — TypeScript · Next.js · RAG / AI Agents

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ABOUT ME

Backend & AI engineer with a Master's in Distributed Systems & AI, focused on building AI-first web applications — RAG knowledge systems, autonomous agents, and robust document/email processing pipelines. Comfortable across the full stack in **TypeScript (Next.js, React, Tailwind, shadcn/ui)** and Python, with a track record of shipping secure, production-grade systems and integrating third-party APIs at scale. I work in an AI-first setup with **Claude / Cursor / Codex** and am looking for a long-term remote role where I can grow with a structured, ambitious team.

WORK EXPERIENCE

Ofoto — Fine Art Digital Studio

Full-Stack & AI Engineer (Final Year Project + Extended Contract) | Feb 2025 – Oct 2025 | Casablanca (remote-friendly)

- **End-to-end AI platform** — designed and shipped a full web application for AI image generation, owning architecture decisions across frontend, backend, and infrastructure.
- **Modern stack & UI** — Vite + TailwindCSS reactive frontend (component-driven, touch-ready), FastAPI + SQLAlchemy + PostgreSQL backend, **Redis-backed async task queues** to handle volume spikes gracefully.
- **3rd-party integrations** — integrated external model APIs (AUTOMATIC1111, InstantID / InsightFace) behind a secured internal gateway, handling **500+ concurrent requests**.
- **Security & delivery** — JWT auth with role-based access (B2C, B2B, Super Admin), CSRF/XSS protection; UML-modeled delivery across 5 sprints via GitHub.
- **Secure cloud deployment** — Docker Compose on Ubuntu/AWS, Cloudflare DNS, Certbot SSL, UFW, WireGuard VPN, fail2ban — **99.9% uptime, -35% latency, -40% time-to-live**.

FSTM Research Laboratory

Research Intern — Attendance Management System | Feb 2023 – Jul 2023 | Mohammedia

- Built a real-time computer-vision pipeline for automated identification against a live database.
- Automated PDF report generation for arbitrary date ranges, removing manual processing.

SELECTED AI PROJECTS

- **RAG knowledge systems** — retrieval pipelines with embeddings, semantic search, and reranking using LangChain/LangGraph and Pinecone vector storage.
- **AI agents & automation** — LLM agents and automated workflows (n8n, Apify, WhatsApp API) to replace repetitive multi-tool processes.
- **Interactive ML platform** — Next.js / TypeScript app turning ML concepts (RAG, transformers, NLP) into interactive, production-quality visualizations.

EDUCATION

ENSET Mohammedia — Master's, Distributed Systems & AI	<i>Bien</i>
FST Mohammedia — Bachelor's, Applied Mathematics	<i>Assez Bien</i>

SKILLS

Languages	TypeScript, Python, Java, SQL
Frontend	Next.js, React, Tailwind CSS, shadcn/ui-style component systems, Vue.js
Backend & APIs	Node/TypeScript services, FastAPI, Spring Boot, REST, gRPC, SQLAlchemy, Redis
AI / LLM Apps	RAG (embeddings, semantic search, reranking), LangChain, LangGraph, LLM agents, vLLM, Ollama, Pinecone, HuggingFace
Data & Pipelines	PostgreSQL, Supabase, MongoDB, ETL, document/email processing, Spark, Airflow, Kafka
DevOps & Cloud	Docker, Kubernetes, AWS, Nginx, GitHub Actions, CI/CD, OAuth2/Keycloak
AI-First Workflow	Claude, Cursor, Codex
ML / DL / CV	PyTorch, TensorFlow, Scikit-learn, Transformers, BERT/DistilBERT, YOLOv8, Stable Diffusion

LANGUAGES

Arabic — Native · English — B1/B2 (solid working proficiency) · French — B1/B2